

Kaushik Bhowmick

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EDUCATION

• Sardar Vallabhbhai National Institute of Technology

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.80

Surat, Gujarat

Dec 2021 – Jun 2025

EXPERIENCE

• Mastercard

Software Developer Engineer I

Pune, Maharashtra

Jun 2025 – Present

- Scaled Mastercard's **Settlement Facilitator** to support European transaction volume growth from 2M toward **60M+ transactions per cycle** by parallelizing PostgreSQL aggregation with **SKIP LOCKED** and isolating aggregate writes per instance — sustaining approximately **5K TPS across 14 systems** with no queue buildup.
- Implemented the **transactional outbox pattern** for Kafka event publication in the Ledger service, guaranteeing at-least-once delivery to downstream consumers and eliminating dual-write inconsistency between the database and message broker.
- Built cross-region transaction consumption to enable **active-active multi-region resiliency**, allowing failover systems to process transactions from any region while retaining home-region ownership of settlement-file generation.
- Built a Java-based **MCP (Model Context Protocol) server** for an internal Kafka platform, automating 6 provisioning workflows and cutting a 20-topic local setup from **30–60 minutes to approximately 2 minutes** — a **15–30× improvement**.

• Atlas Consolidated Pvt. Ltd.

Software Developer Engineer Intern

Hyderabad, Telangana

Jan 2025 – May 2025

- Re-architected a **Kafka-based hierarchical aggregation pipeline** handling 1M+ financial transactions, replacing recursive parent-node updates with **indexed, prefix-based aggregation** over leaf accounts to remove processing bottlenecks and preserve balance consistency.
- Built dynamic JSON-to-Protobuf mapping using **Protocol Buffers Struct** for a secure inter-service proxy API (POST/PUT/DELETE), eliminating the need for route-specific Protobuf schema definitions.
- Implemented **bulk-upload validation** and a state machine enforcing 3 accounting-book lifecycle states, with an extensible rejection-reason model for handling suspense transactions.

• Mastercard

Software Developer Engineer Intern

Pune, Maharashtra

May 2024 – Jul 2024

- Built a scheduler-tracking service in Java/Spring Boot with a **scheduler-agnostic interface contract**, providing unified execution monitoring across Spring Scheduler and Quartz — allowing new scheduler configurations to integrate without modifying core tracking logic.

PROJECTS

• Aurora Chess Engine (GitHub)

Technologies: Modern C++

- Built a UCI-compatible C++20 chess engine with bitboards, PEXT/BMI2 attack tables, alpha-beta/PVS, quiescence search, Zobrist transposition tables, and **AVX2-optimized NNUE evaluation**; generated **119M depth-6 nodes at 18.3M NPS**.
- Deployed the engine on Lichess as PurpleHunter via UCI-based bot integration, completing **8,000+ rated games**.

• Relaxed Graph Substitutions Using ONNX (GitHub)

Technologies: Python, ONNX, TensorFlow

- Implemented a non-greedy ONNX graph optimizer using cost-bounded backtracking to explore relaxed substitutions beyond locally optimal transformations.
- Reduced benchmarked inference latency by **36.8%**, from **706 μs to 446 μs**, while reducing operator count from **4 to 3** through a 3-step substitution sequence.
- Created a hardware-agnostic static cost model combining **3 factors** — operator computation, memory-access latency, and model parallelism — to rank candidate graph transformations without executing each model.

TECHNICAL SKILLS

Languages

Java, C++, SQL, TypeScript, Python

Backend and Frameworks

Spring Boot, Spring Scheduler, Quartz, gRPC, Protocol Buffers, REST APIs

Distributed Systems

Kafka, Transactional Outbox Pattern, Concurrency Control, Multi-Region Resiliency, Event-Driven Architecture

Databases

PostgreSQL, MySQL, Redis, Maxwell CDC

Testing and Tools

JUnit, Mockito, Git, Linux, Docker, CMake

Cloud

AWS (EC2, RDS, SQS, S3)